

## **CHAPTER 2**

### **SURVEY OF TECHNOLOGIES**

#### **2.1 SURVEY OF TECHNOLOGY**

##### **2.1.1 FRONTEND**

The frontend of the Digital Fundraising Portal System is essential for delivering a user-friendly and interactive experience. It enables project creators and supporters to engage with the platform effectively, whether creating campaigns, viewing funding progress, or making contributions.

#### **HTML (HYPERTEXT MARKUP LANGUAGE)**

- HTML structures the content and layout of the web pages, serving as the foundation for the user interface.
- It allows developers to organize content using elements like headings, forms, buttons, tables, and sections.
- HTML ensures the content is accessible and compatible across all browsers, offering a consistent experience.

#### **ADVANTAGES:**

- Simple and easy to learn
- Compatible with all browsers
- Provides a semantic and organized content structure
- Integrates easily with CSS and JavaScript

#### **DISADVANTAGES:**

- Static by default; cannot handle dynamic behavior alone
- Depends on CSS and JavaScript for styling and interactivity
- Limited functionality for complex user interfaces

## **CSS (CASCADING STYLE SHEETS)**

- CSS styles the visual components of the application, including layout, colors, fonts, and responsiveness.
- It separates presentation from structure, promoting cleaner code and reusability across multiple pages.
- With media queries, CSS ensures the application is fully responsive and adapts well to both desktop and mobile devices.

### **ADVANTAGES:**

- Enables consistent styling across all pages
- Supports responsive design through media queries
- Reduces redundancy by using external stylesheets
- Easy to update the look and feel of the site

### **DISADVANTAGES:**

- Can become complex to manage in large projects without proper structure
- Browser compatibility issues may arise
- No built-in logic (unlike JavaScript)

## **TYPESCRIPT**

- TypeScript adds interactivity to the platform, handling client-side validation, live content updates, and user interactions.
- It plays a crucial role in features like live funding progress bars, real-time feedback, and responsive design.
- TypeScript operates within the browser and allows the development of a seamless, dynamic user experience.

## **ADVANTAGES:**

- Adds type safety to JavaScript, reducing runtime errors
- Improves code readability and maintainability in large projects
- Supports object-oriented programming concepts such as interfaces, enums, and generics.
- Seamlessly integrates with modern JavaScript libraries.
- Compiles to clean, browser-compatible JavaScript

## **DISADVANTAGES:**

- Requires a compilation step, adding complexity to the build process
- Slightly steeper learning curve for beginners unfamiliar with typed language.

### **2.1.2 BACKEND – SUPABASE**

Supabase serves as the backend-as-a-service (BaaS) platform for the Digital Fundraising Portal System. It provides a powerful, scalable alternative to traditional backend development by combining a PostgreSQL database with built-in authentication, storage, and real-time subscriptions.

## **ADVANTAGES:**

- Easy Setup & Scalable: Supabase provides an instant backend with database, auth, and APIs, reducing development time.
- Built-in Auth & RLS: Includes secure user authentication and powerful row-level security policies.
- Realtime Features: Allows live syncing of data changes (e.g., new bookings, availability updates).

## **DISADVANTAGES:**

- Limited Customization vs. Traditional Backend: Some constraints when compared to building a full custom backend from scratch.
- Learning Curve for Policies: Row-level security can be tricky to configure for beginners.

### **2.1.3 DATABASE – POSTGRESQL**

The database layer is managed by Supabase, which uses PostgreSQL, a robust and scalable open-source relational database system.

#### **POSTGRESQL**

- PostgreSQL handles the storage of users, projects, contributions, reward tiers, and transaction data.
- It supports complex queries, data relationships, and transactions, which are vital for maintaining data integrity in financial systems.
- PostgreSQL's support for JSON, indexing, and triggers allows flexibility in storing structured and semi-structured data efficiently.
- With Supabase's interface, PostgreSQL is managed easily through a web dashboard, SQL editor, and API layer.

#### **ADVANTAGES:**

- Fully ACID-compliant, ensuring data integrity and consistency
- Supports complex queries, joins, triggers, and stored procedures
- Allows relational data modeling with strong support for foreign keys and constraints
- Highly extensible with custom functions, data types, and full-text search

#### **DISADVANTAGES:**

- Slightly more complex to configure and optimize than simpler databases like SQLite or MySQL
- Requires knowledge of SQL for effective querying and database management
- Can consume more system resources compared to lighter databases